On the Inaccuracy of Global Optimization Methods to Extract ReRAM Parameters

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Abstract

We fit the Dynamic Memdiode Model (DMM) of Miranda et al. to a 14 604-cycle HfO_2 ReRAM switching-current dataset, using five distinct parameter-extraction strategies: gradient-based least squares, a genetic algorithm, a directed genetic algorithm steered by the local Jacobian's principal directions, a genetic-algorithm/least-squares hybrid, and a neural-network surrogate trained on realistic synthetic data. All five reproduce the measured current to within a few percent. None of them reliably recovers the underlying twelve model parameters. We show this with a Monte Carlo test against a known, noise-free synthetic ground truth: even there, several parameters – most persistently the SET threshold voltage v_s, the SET nonlinearity pair (a_{on}, i_{on}), and the RESET rate constant η_r – are recovered with correlations to truth no better than 0.1–0.3, regardless of optimizer. We trace this to a structural, not algorithmic, cause: every cycle in the dataset is excited by the same fixed voltage waveform, which does not carry enough independent information to pin down every parameter of a twelve-parameter nonlinear ODE. We corroborate this directly with a new test reported here: pooling a_{on}, i_{on} as shared constants across twelve real measurement windows, instead of fitting them per window, multiplies the fit error by a factor of 14.6 uniformly across every window, ruling out the simplest fix one would try. A sixth, categorically different method – a non-gradient, rule-based extractor with no joint cost function at all – fails the same benchmark too, correlating weakly or negatively with truth on every one of its ten parameters (average -0.04, the worst of the six), which we read as further, optimizer-independent evidence for the same conclusion. We also describe the subset of extraction relations that *can* be derived directly from the model's own equations and validated exactly, the production fitting procedure used to generate the dataset-wide results, and release the parameter-trajectory viewer used throughout this work as a public tool.

1 Introduction

Compact modeling of resistive switching devices has mostly been judged by how well a model reproduces the major I-V hysteresis loop. This is a weak test: a model with more free parameters than the data can constrain will often match the loop closely while assigning those parameters values that have little to do with the device's actual physics. Hysteron behavior – partial, minor loops traced out under sub-saturating excitation – has recently been proposed as a more stringent benchmark, since it probes the device away from its two extreme states and is harder to fit by accident [1]. The Dynamic Memdiode Model (DMM) we use in this work is one such model, and it is the one against which this paper's findings are demonstrated; but the problem we describe is not specific to it. It belongs to parameter extraction for resistive-switching compact models generally, and shows up, in one form or another, in essentially every model we could find in the literature that documents how its parameters are actually obtained from data.

The Stanford/ASU RRAM model [2] is calibrated against experimental I-V sweeps by nonlinear optimization, with cycle-to-cycle variation captured afterwards as a distribution over the fitted parameters rather than derived from first principles. The Simmons tunnel-barrier model of Pickett et al. [3], one of the earliest physically motivated switching-dynamics models, has several constants that later authors describe plainly as fitting parameters without direct physical meaning, extracted by matching switching data rather than read off a curve. The TEAM/VTEAM family [4] was designed with extraction in mind – its threshold voltages are meant to correspond to visible features of the I-V curve – but even there the full parameter set is obtained by numerical fitting, not a closed-form procedure. A 2025 preprint by Hamid and Hassan [5] tackles the Stanford and VTEAM models with almost exactly the architecture we independently arrived at for the DMM below: a convolutional network trained on synthetic data produces an initial guess, refined afterwards by heuristic optimization. Their own results report several parameters saturating at the imposed bounds of the search space during fitting – the same symptom, on a different model, that we document in Section 6 for the DMM's a_{on}. One genuinely different approach is the data-driven Verilog-A model of Messaris et al. [6], which pairs the model with dedicated characterization hardware that applies a chosen sequence of pulses to the physical device and extracts parameters from the elicited response; this sidesteps the identifiability problem by controlling the excitation rather than by fitting after the fact, at the cost of requiring live access to the device under a purpose-built protocol, which is not available for the dataset used here.

Closer to home, the DMM's own origin group had already met this problem. Aguirre, Miranda and coauthors [7] replace the iterative optimizer entirely with a convolutional network trained on synthetic DMM curves, reporting roughly an order-of-magnitude speed-up over conventional fitting on large datasets. Speed, however, is not accuracy, and that paper does not report a ground-truth recovery test of the kind we run in Section 5; our own neural surrogate (Section 3) is built in the same spirit, and we show directly where it helps and where it does not.

The identifiability literature gives a vocabulary for what we observe. A model is *sloppy* when its cost function has a wide spread of curvatures around the optimum: a few directions in parameter space are tightly constrained by the data, and most are not, forming long, flat valleys that an optimizer can wander along at essentially no cost in fit quality. Chis and Balsa-Canto [8] make the point, often missed, that sloppiness and non-identifiability are not the same thing – a sloppy parameter combination can still be identifiable in principle, given the right experiment – while White et al. [9] warn that the usual remedy, optimal experimental design, is itself fragile in sloppy systems and can fail to deliver what it promises. Both results are directly relevant to what follows: the DMM parameters we cannot recover are not poorly optimized, they are poorly constrained by the *specific, single, repeated* excitation waveform in this dataset, which is exactly the situation this literature describes.

Finally, one of the methods we test here, the directed genetic algorithm of Section 3, is a reimplementation of our own earlier work on parameter extraction for organic thin-film transistor models [10], adapted from that context to the DMM's twelve-parameter chain-fitting problem.

The remainder of this paper is organized as follows. Section 2 summarizes the DMM equations. Section 3 describes our production fitting procedure and the five extraction strategies compared. Section 4 collects the extraction relations we can derive and validate directly from the model's own equations, as opposed to fitting by brute-force optimization. Section 5 reports the Monte Carlo ground-truth comparison across all five methods. Section 6 works through why fitting this model is hard, closing with a new pooled-fitting test that rules out the most obvious candidate fix. Section 7 concludes. Appendix A describes the rolling-parameter analyzer used to produce most of the figures in this paper, released as a public repository.

2 The Dynamic Memdiode Model

The DMM, as formulated by Miranda et al. [1], separates the device response into a fast, algebraic current-voltage relation and a slow, first-order ODE describing an internal state variable $g \in [0, 1]$ (0: fully OFF, 1: fully ON). Writing $K(a, b, g) = (a - b)g + b$ for the model's one shared endpoint-interpolation rule,

$$I(V_A, g) = K(i_{on}, i_{off}, g)\,\sinh\big(K(a_{on}, a_{off}, g)\,V_A\big), \tag{1}$$

$$V = V_A + K(r_{on}, r_{off}, g)\,I(V_A, g), \tag{2}$$

where V is the terminal voltage and V_A the internal memdiode voltage, related by the drop across a state-dependent series resistance. The state evolves as

$$\frac{dg}{dt} = (1 - g)\,S(V_A)\,H(V_A)\,H(dV/dt) \;-\; g\,R(V_A)\,H(-V_A), \tag{3}$$

$$S(V) = 2\,e^{-\eta_s v_s}\sinh(\eta_s V), \tag{4}$$

$$R(V) = 2\,e^{\eta_r v_r}\sinh\big(-\eta_r\,g^{\gamma}V\big), \tag{5}$$

with $H(\cdot)$ the Heaviside step. SET kinetics are gated by $V_A \geq 0$ *and* a rising terminal voltage; RESET kinetics are gated only by $V_A < 0$, with no dependence on sweep direction. This asymmetry is not a detail: it is the reason, derived and confirmed in Section 4, that the two turning points of a sweep behave differently under a finite-difference analysis.

Twelve parameters describe the device, listed in Table 1 together with which half-cycle each one is actually free to vary in during a chained fit – a grouping we refer to throughout as the SET side, $\{v_s, \eta_s, r_{on}, a_{on}, i_{on}\}$, and the RESET side, $\{v_r, \eta_r, \gamma, r_{off}, a_{off}, i_{off}\}$.

Table 1: The DMM's twelve parameters: what each one controls, and which half-cycle it is free to vary in during a chained fit (Section 3).

parameter	description	free on
r_{on}	series resistance at $g = 1$ (fully ON)	SET
r_{off}	series resistance at $g = 0$ (fully OFF)	RESET
a_{on}	sinh argument scale (nonlinearity) at $g = 1$	SET
a_{off}	sinh argument scale (nonlinearity) at $g = 0$	RESET
i_{on}	current scale at $g = 1$	SET
i_{off}	current scale at $g = 0$	RESET
v_s	SET threshold voltage	SET
v_r	RESET threshold voltage	RESET
η_s	SET switching rate constant	SET
η_r	RESET switching rate constant	RESET
γ	RESET's state-dependence exponent (g^{γ} in $R(V)$)	RESET
g_0	initial internal state, at the start of the fitted stretch	both (shared)

g_0 is the one exception to the SET/RESET split: it is not re-fit on every half-cycle the way the other eleven are, since, as Section 3 describes, g itself is carried forward continuously rather than reset – g_0 only ever matters as the starting condition for the very first window of a fitted stretch, after which the chain's own trajectory takes over.

Why chain the fit across cycles at all, rather than fit each cycle on its own? Because the measurement itself gives no license to do otherwise: it is a single, continuous physical trace, not a sequence of independent snapshots. The real device enters every half-cycle in whatever internal state the previous one left it in – not reset to a fresh, known condition – and g's own governing equation, Eq. (3), is a first-order ODE with no discontinuous reset written into it anywhere; it can only evolve continuously, driven by whatever $V_A(t)$ and the gating terms dictate at each instant, as a continuation of one physical process, not a new one starting fresh at every cycle boundary. Fitting each cycle independently – restarting every window from the same fixed guess, blind to what the device was doing one half-cycle earlier – would tacitly assume the opposite: that the state, and by extension at least some of the parameters that govern it, are free to jump discontinuously at a boundary the measurement itself does not mark as special in any way. Nothing supports that assumption, and for g it is provably false. This is the basic motivation for the chain: g must carry over exactly from one half-cycle to the next, and our production procedure extends the same continuity expectation, more cautiously, to the other side's own just-fit parameters as well (held fixed rather than re-guessed), on the grounds that whatever physical mechanism they represent is equally unlikely to jump at a boundary the experiment does not distinguish. An independent per-cycle fit discards this information entirely and re-derives each window's full twelve-parameter state from the window's own data alone – already a weakly constrained problem on a single window, as Section 6 documents in detail; throwing away the one piece of extra information the experiment actually offers for free (continuity with its own immediate past) only makes that problem harder, not easier.

This grouping matters because, as described in full in Section 3, the two sides are never fit together: a given half-cycle updates only its own side's free parameters, with the other side held fixed at whatever value it was last fit to. Write S_k for the SET side after its k-th update and R_k for the RESET side after its k-th update ($k = 0$ denoting whatever was inherited before the stretch of data under consideration). Figure 1 shows two real, consecutive full cycles with the resulting sequence of active parameter pairs marked directly on the data: the first SET half-cycle ($0 \to V_{max} \to 0$) is simulated and fit with R_0 held fixed, producing S_1, so the pair actually used for that window is (S_1, R_0); the RESET half-cycle that follows holds S_1 fixed and produces R_1, giving (S_1, R_1); the next cycle's SET half-cycle holds R_1 fixed and produces S_2, giving (S_2, R_1); and so on,

$$(S_1, R_0) \;\to\; (S_1, R_1) \;\to\; (S_2, R_1) \;\to\; (S_2, R_2) \;\to\; \cdots, \tag{6}$$

one index advancing at a time, never both at once. g itself is not part of either pair: it is not reset at any of these boundaries, but carried forward continuously as the true, single state trajectory underlying all four windows in the figure, exactly as Section 4.1's discussion of incomplete switching requires. We show this explicitly because it is easy to misread a chain-fitted CSV as one 12-parameter vector per cycle, when it is really this alternating, half-cycle-resolved sequence – a confusion that, left unresolved, previously produced a spurious reconstruction error in the viewer described in Appendix A.

3 Fitting procedure

3.1 Production chain fit

The dataset is a real measurement on a TiN/Ti/HfO_2/W device from the same research group and measurement line as [11], a sequence of triangular voltage sweeps, each cycle tracing $0 \to V_{max} \to 0 \to V_{min} \to 0$. We split each cycle into its two half-cycles – $0 \to V_{max} \to 0$ (SET side

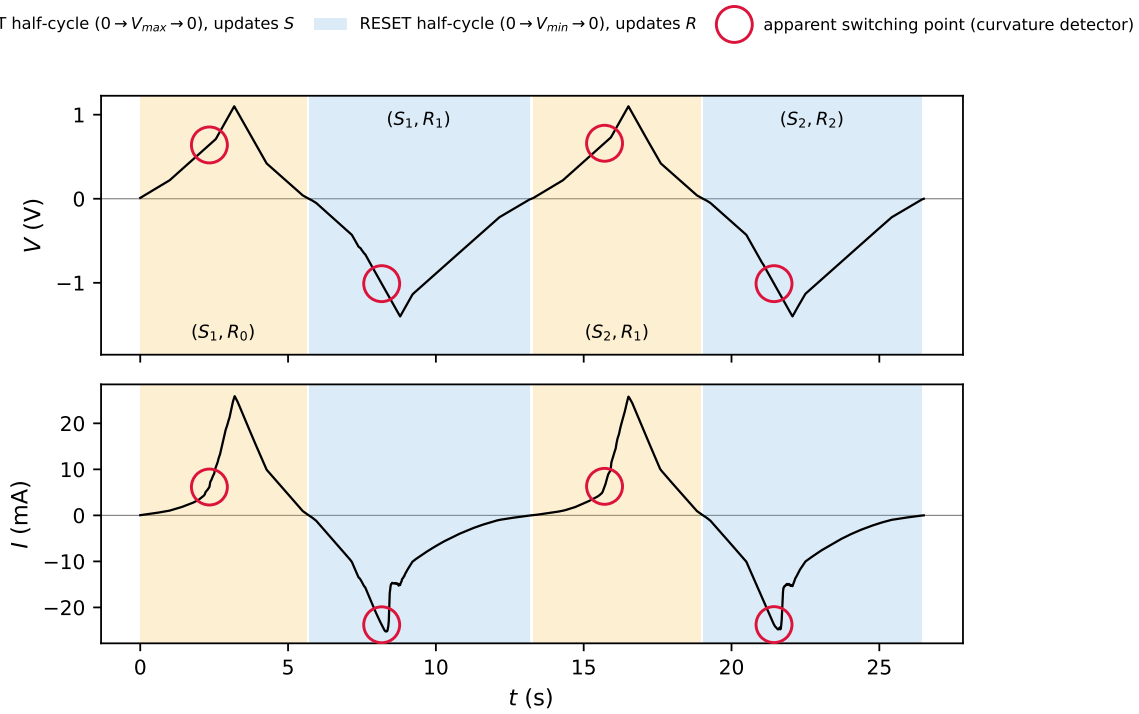


Figure 1: Two real, consecutive full cycles (3000–3001): applied voltage (top) and measured current (bottom), with the four half-cycle spans shaded and labeled by the parameter pair the chain fit actually uses for that window, following Eq. (6). Only one side's index advances per half-cycle; the other side's value carries over unchanged from the previous window. The open circles are a loose, approximate marker of where each half-cycle's own apparent switching point sits, located by the same curvature-based detectors described in Section 4 (d^2I/dt^2 for SET, d^3I/dt^3-before-d^2I/dt^2 for RESET) – shown here to make visible, on real data, roughly where within each shaded span the transition driving that span's own parameter update actually happens, not as a precise measurement of v_s or v_r.

active) and $0 \to V_{min} \to 0$ (RESET side active) – found from the raw voltage's own zero crossings, with no detection logic beyond that. Each half-cycle is fit independently by a joint, simultaneous least-squares call over its own five (SET) or six (RESET) free parameters, with the other side's parameters held fixed at their value from the immediately preceding half-cycle and the internal state g carried forward exactly (the ending g of one half-cycle becomes the starting g of the next, never reset). The residual vector includes both the current $I(t)$ and, by default, a smoothed dI/dt, since matching the current alone without penalizing its derivative was found early on to admit visibly wrong fits with an unremarkable aggregate error.

Three safeguards, each added after a specific documented failure mode, make this procedure usable across the full dataset rather than a few hundred cycles at a time:

- **A pin-escape margin.** Any of the twelve parameters that drifts within 5% (in log space, for the log-scaled parameters) of its own global bound is reset to a geometric-mean value before the next half-cycle, to stop a parameter from ratcheting toward a bound and never returning.
- **A capped η_s, η_r range and a fixed γ.** Left free, γ and η_r trade off against each other: across roughly a threefold range of γ, a compensating η_r reproduces the data to within 0.2–

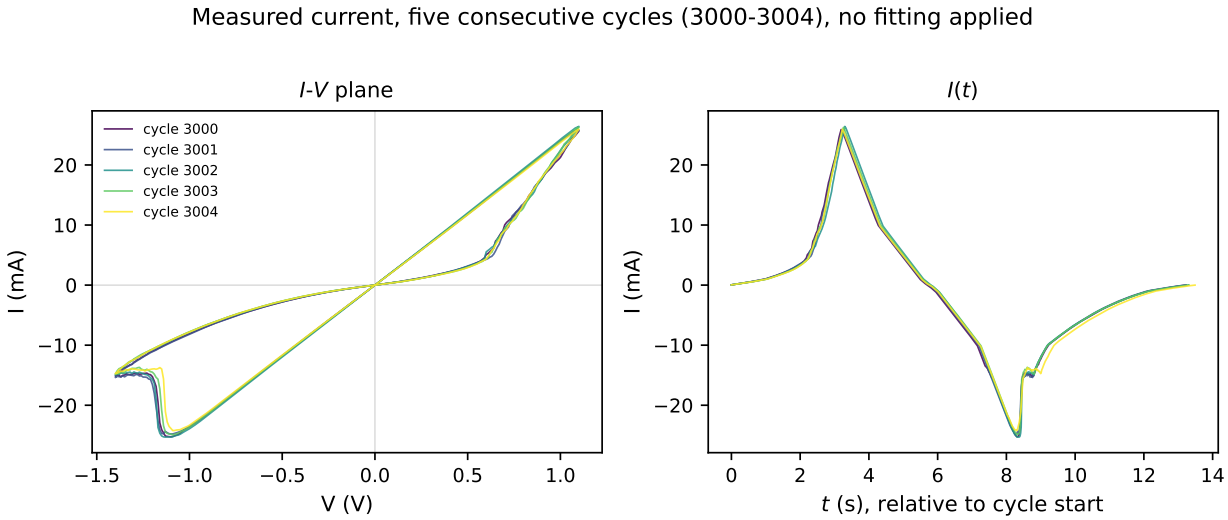


Figure 2: Measured current, five consecutive raw cycles (3000–3004), no fitting applied. Left: the major I-V hysteresis loop. Right: the same five cycles' current against time, relative to each cycle's own start. Both the loop shape and the time trace are stable from cycle to cycle, which is precisely what makes the waveform-diversity argument of Section 6 bite: every cycle probes essentially the same region of (V, g) space, at essentially the same rate.

5.1% error, a textbook flat ridge in the joint objective, not a sharp minimum. Fixing $\gamma = 0.22$ and capping $\eta_s, \eta_r \leq 300$ removes this ridge; both choices are validated against the model's own cost surface, not asserted by convention.

- **An rmse-spike retry.** A half-cycle whose fit error exceeds three times a rolling baseline (the median of the preceding 50 half-cycles) is refit once from a geometric-mean restart rather than from the possibly already-bad carried-forward value, keeping whichever attempt fits better. This catches a specific, otherwise fatal failure mode: a half-cycle whose own applied voltage range cannot physically reach its carried-forward v_s or v_r sits in an exactly flat region of the cost surface (we measured bit-for-bit identical residuals for v_r perturbed by ± 0.1 V around the stuck value), where a gradient-based optimizer has no local signal to climb out on and simply returns its starting point. A single such half-cycle, unpatched, derailed 69% of a full 16 602-cycle run and never recovered.

With all three in place, a full 14 604-cycle run (cycles 2000–16603) gives a SET-side peak-current error of 0.7% (median) with only 0.09% of half-cycles exceeding 10% error, and a RESET-side peak-current error of 1.7% (median, 5.35% of half-cycles above 10%); the fit never freezes for more than a handful of consecutive half-cycles anywhere in the run. Figure 3 shows one such window: the fit tracks the measured current closely through the transition, with the largest local deviation visible on the rising edge just before the peak, a reminder that "low aggregate rmse" and "correct everywhere" are not the same statement. Figure 2 shows five consecutive raw measured cycles, to give a sense of what this dataset's major loop and its cycle-to-cycle repeatability actually look like before any fitting is applied.

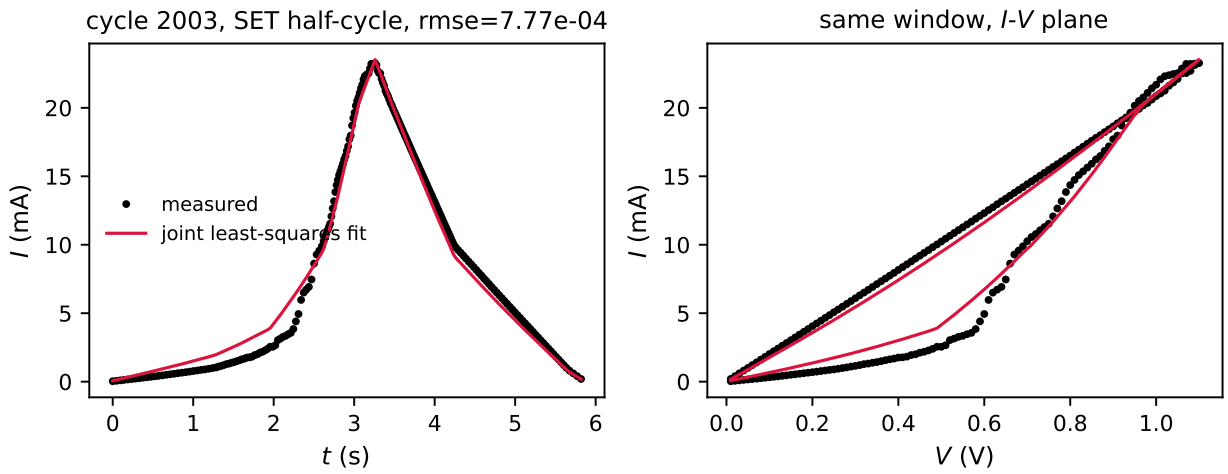


Figure 3: A real SET half-cycle window (cycle 2003) fit by the joint least-squares procedure. Left: measured and fitted current against time. Right: the same data in the I-V plane. The fit is close but not exact; the largest local deviation sits on the rising edge, just before the current peak.

3.2 Five extraction strategies

We compare five ways of solving each half-cycle's local optimization problem, all sharing the chain-level procedure above:

lsq `scipy.optimize.least_squares` (trust-region-reflective), warm-started from the previous half-cycle's own fit. The fastest method tested (roughly 5–25 half-cycles/s depending on configuration) and, on both real data and the synthetic ground truth of Section 5, the most accurate on average.

GA `scipy.optimize.differential_evolution`, with whichever of v_s/v_r is this half-cycle's own free threshold locked to a narrow window around a direct, curvature-based estimate of that threshold (Section 4), so the population search does not wander far from a value the data itself weakly constrains. Roughly 20–40 times slower than lsq.

directed GA Our own earlier method [10], reimplemented for this model: each generation, the residual Jacobian at the current best point is used to build the Gauss–Newton approximation $J^\top J$, whose eigendecomposition gives a local principal-axis frame; new individuals are Gaussian steps in that frame, with step size scaled by $\sqrt{\lambda_i/\lambda_{max}}$ so the stiffest directions are searched more aggressively, mixed with a 30% isotropic-random fraction so the population cannot get trapped crawling a sloppy valley floor. Roughly 5.7 times faster than plain GA at comparable fit error.

GA+lsq A multi-start hybrid: the full final population of a differential-evolution run, not just its best member, is individually polished by a bounded local least-squares step, and the best polished candidate is kept. The most expensive method tested, at roughly 17 times lsq's cost.

CNN+lsq A feed-forward network (not a true 1-D convolutional architecture in the run reported here, a limitation we return to in Section 6) trained to map a half-cycle's resampled current and voltage traces, together with the fixed side's parameters and g_0, directly to the free

side's parameters, trained on synthetic curves generated from realistic parameter combinations rather than a uniform prior over the box bounds – a uniform prior was tried first and gave a training set dominated by curve shapes nothing like a real device's fit, which produced a catastrophic standalone evaluation. The network's output seeds one local least-squares polish before being reported.

4 Extraction relations derived from the model's own equations

Alongside the five optimization-based strategies above, the DMM's own algebra yields several relations that can be checked exactly, in closed form, against the model equations of Section 2, rather than being validated only empirically. We derive the general result they all specialize from, then list each one together with why it holds, since this project also tried several plausible-looking shortcuts that turned out *not* to hold, and the distinction matters for anyone reusing this methodology.

4.1 General decomposition of dI/dt

Write $K_a \equiv K(a_{on}, a_{off}, g)$, $K_i \equiv K(i_{on}, i_{off}, g)$, $R_s \equiv K(r_{on}, r_{off}, g)$ for the three state-dependent blends of Section 2, each affine in g with slope $(a_{on} - a_{off})$, $(i_{on} - i_{off})$, $(r_{on} - r_{off})$ respectively. From $I(V_A, g) = K_i \sinh(K_a V_A)$, the two partial derivatives are

$$J \equiv \left.\frac{\partial I}{\partial V_A}\right|_g = K_i K_a \cosh(K_a V_A), \tag{7}$$

$$\left.\frac{\partial I}{\partial g}\right|_{V_A} = (i_{on} - i_{off}) \sinh(K_a V_A) + K_i V_A \cosh(K_a V_A)(a_{on} - a_{off}). \tag{8}$$

Differentiating the series relation $V = V_A + R_s(g)\, I(V_A, g)$ with respect to t along a trajectory $(V_A(t), g(t))$ gives

$$\frac{dV}{dt} = \big(1 + R_s J\big)\frac{dV_A}{dt} + \underbrace{\Big[(r_{on} - r_{off})\, I + R_s \frac{\partial I}{\partial g}\Big]}_{C} \frac{dg}{dt}, \tag{9}$$

which rearranges to $dV_A/dt = \big(dV/dt - C\, dg/dt\big)/(1 + R_s J)$. Substituting into $dI/dt = J\, dV_A/dt + (\partial I/\partial g)\, dg/dt$ and collecting terms gives the exact decomposition

$$\boxed{\frac{dI}{dt} = A(V, g)\, \frac{dV}{dt} + B(V, g)\, \frac{dg}{dt}}, \qquad A \equiv \frac{J}{1 + R_s J}, \qquad B \equiv \frac{\partial I}{\partial g} - A\, C. \tag{10}$$

This is exact, not a small-signal approximation; it is an identity given the model's own equations, true at every sample of a trajectory the forward model has already produced. Used that way, on a trajectory from `dmm.simulate()`, it is a diagnostic: a way to check a candidate fit's own internal consistency. Its use as an *extraction* tool, on raw data where $V_A(t)$ and $g(t)$ are not known, comes instead from three special cases below where the slow term either vanishes or can be cancelled, leaving only quantities measurable directly from (t, V, I).

A diagnostic the decomposition gives for free: incomplete switching at the turning point. Apply Eq. (10) on each side of V_{max} separately, normalizing by the ramp rate r so the sweep's own direction reversal ($dV/dt = +r$ before, $-r$ after) is divided out and does not by itself

masquerade as a slope change. SET's hard gate means $dg/dt|_{t_0^-} = (1-g)S(V_A)$ immediately before the turn and $dg/dt|_{t_0^+} = 0$ immediately after, so

$$\left.\frac{dI}{dV}\right|_{t_0^-} - \left.\frac{dI}{dV}\right|_{t_0^+} = \frac{B\,(1-g)\,S(V_A)}{r}. \tag{11}$$

This vanishes if and only if $(1-g) = 0$. In other words, a measurable kink in the I-V loop right at V_{max} – the pre-turn and post-turn slopes visibly disagreeing, rather than the loop turning smoothly – is not a measurement artifact or a modeling inconvenience: it is direct, exact evidence, from the model's own equations, that g had *not* reached 1 by the time the sweep reversed, i.e. that SET was still incomplete at V_{max}. Had the device fully saturated before the sweep turned around, $(1-g)$ would already be zero on the pre-turn side as well, Eq. (11) would vanish, and the loop would turn smoothly. The real loops in Figure 2 and the fitted window in Figure 3 both show a visible, non-smooth corner at their own turning points, consistent with g sitting strictly inside $(0, 1)$ there rather than at either extreme – not a special or occasional occurrence in this dataset, but the typical case. (The symmetric argument at V_{min} does not hold the same way, since RESET is not sweep-direction gated and dg/dt does not jump there; this diagnostic is specific to the SET side's hard gate.)

We flag this as an *added* difficulty for extraction, on top of the parameter non-identifiability discussed in Section 6: it means the internal state cannot be treated, even approximately, as "reset" to a known boundary value (0 or 1) at either end of a sweep – the state the device is actually in at any turning point is itself an unknown, interior value, set by the full prior trajectory (how far switching had already progressed when the previous half-cycle ended, how much of the current half-cycle's finite duration was available for the present transition to run, and the sweep's own rate). This is precisely why the production chain fit of Section 3 carries g continuously across half-cycles, via `g0_override`, rather than resetting it: no cycle in this dataset can be fit correctly in isolation, starting from an assumed fully-off or fully-on state, and the degree of completeness at each turning point is one more quantity entangled with the already weakly-identified rate parameters η_s, η_r rather than a simplifying boundary condition we can take for granted.

4.2 Near-$V = 0$ series resistance

Deep inside a SET or RESET plateau, far from any transition, g is effectively frozen ($dg/dt \approx 0$) and $V_A \approx 0$. At $dg/dt = 0$, Eq. (10) reduces to $dI/dt = A\,dV/dt$, i.e. $dI/dV = A$; evaluating A at $V_A \approx 0$, where $J = K_i K_a \cosh(0) = K_i K_a$ from Eq. (7), gives

$$\left.\frac{dI}{dV}\right|_{V\approx 0} = A|_{V_A=0} = \frac{K_a K_i}{1 + R_s K_a K_i}, \qquad \text{equivalently} \qquad \frac{1}{\text{slope}_0} = R_s + \frac{1}{K_a K_i}, \tag{12}$$

confirmed numerically exact against the forward model, and a direct corollary of the general decomposition rather than a separate result. This lets R_s be solved in closed form given a measured near-zero slope and a trial (K_a, K_i) – the basis of `derive_ron`/`derive_roff` – or, with no trial values at all, a crude but useful seed $R_s \approx 0.5/\text{slope}_0$ (`estimate_rs_half`), motivated by an independent check that this 50/50 split between the series resistance and the fast element's own resistance holds on average across real SET windows. The relation is provably exact on the ON side, which has a genuine frozen-g segment near $V = 0$; it does not transfer to the OFF side, which never fully freezes for $V < 0$ in this model – a real identifiability limit of a single window's data, not an implementation gap.

4.3 Turning-point asymmetry

At any sweep turning point, $dV/dt = 0$ by definition, so Eq. (10) collapses to $dI/dt = B\,dg/dt$: the slow term alone, exactly, with no approximation. What is *not* symmetric is what happens on either side of that point, because of the gating in Section 2's own dg/dt equation: at V_{min}, RESET kinetics ($dg/dt = -gR(V_A)$) are indifferent to sweep direction, so a finite-difference estimate of A built from samples on either side of the turning point is unbiased either way. At V_{max}, SET kinetics switch off *exactly* at the peak ($H(dV/dt)$ is a hard gate, not a smooth one), so dg/dt jumps from $(1-g)S(V_A)$ to precisely zero the instant the sweep turns, rather than merely changing sign. Immediately after the peak, Eq. (10) therefore reduces further, to $dI/dt|_{t_0^+} = A\,dV/dt|_{t_0^+}$ with no slow-term contamination at all – a clean, single-sided read of A – while the pre-peak sample still carries a live, nonzero $B\,dg/dt$ term that a symmetric two-sided estimate would wrongly average in. The practical consequence, confirmed against a synthetic ground truth over 600 real half-cycle windows: the post-peak slope *alone* gives a median A error of 0.13% at V_{max}, while averaging in the pre-peak sample – the natural thing to do for a symmetric finite difference – actively hurts (57% median error), exactly as the asymmetric gating above predicts. At V_{min}, by contrast, all finite-difference variants agree to within a few percent, exactly as the symmetric derivation predicts.

4.4 Mid-sweep kinks

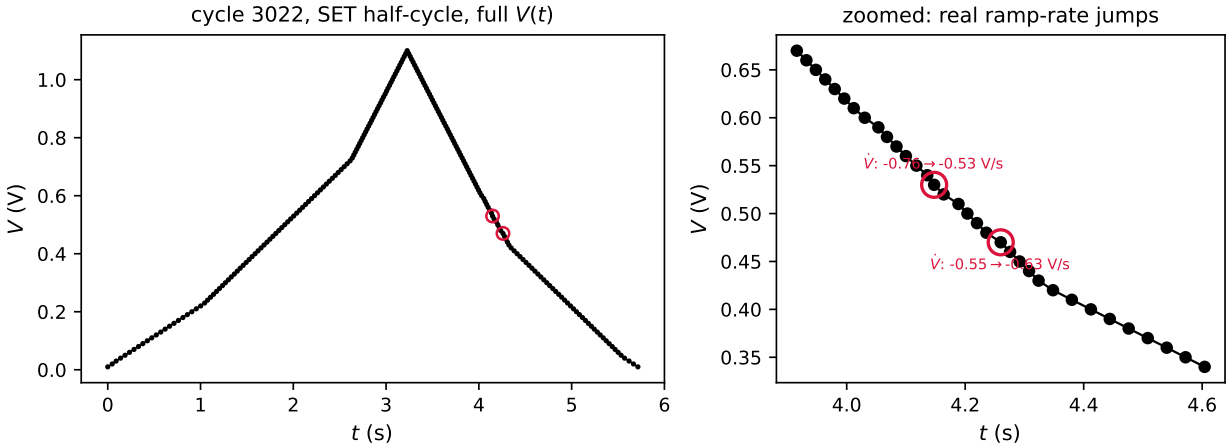


Figure 4: The real applied-voltage sweep is not a constant-rate ramp. Left: one full real SET half-cycle's $V(t)$ (cycle 3022), with the ramp-rate changes visible as genuine corners, not smoothing artifacts. Right: a zoomed-in stretch of the same curve, with two ramp-rate jumps located by `find_mid_sweep_kinks()` (same detector and default thresholds as the validation below), each labeled with its own measured $\dot{V}$ before and after.

The real instrument does not ramp at a constant rate; it holds one dV/dt for a run of samples and then jumps to another, visibly so, as Figure 4 shows directly on real data: the left panel is one full real half-cycle's $V(t)$, with a piecewise-linear, not smoothly curved, shape even at this scale; the right panel zooms into a short stretch of it and marks two genuine ramp-rate jumps found by the same detector described below, each with its own before/after $\dot{V}$. Across such a jump, if g's own gate state does not change and dg/dt is locally constant (not zero, merely unchanging across

the two samples), Eq. (10) gives, for the difference between the two sides of the jump,

$$\Delta\left(\frac{dI}{dt}\right) \approx A\,\Delta\left(\frac{dV}{dt}\right) \quad \Longrightarrow \quad A_{\text{apparent}} = \frac{\Delta(dI/dt)}{\Delta(dV/dt)}, \tag{13}$$

the $B\,dg/dt$ term having cancelled in the difference rather than vanished outright – the same mechanism as the turning-point case, applied as a two-point finite difference instead of a one-sided limit, and usable anywhere in the sweep where the raw instrument happens to change its own ramp rate, not only at V_{max}/V_{min}. We use two filters before trusting a candidate jump: dV/dt itself must be near-constant on *both* neighboring samples (rejecting multi-sample instrument transients that would otherwise straddle a continuous ramp rather than two genuine plateaus), and the jump must not sit too close to the window's own peak voltage (checked per half-cycle piece, since a whole-cycle check under-filters whichever side has the smaller peak magnitude). Validated over 200+ real half-cycles against synthetic ground truth: median A error around 7%, with the outliers explained almost entirely by $|dg/dt|$ itself – outlier kinks sit at a median $|dg/dt| \approx 0.42$, clean ones at $\approx 1.2 \times 10^{-5}$ – i.e. Eq. (13) fails specifically where its own "locally constant dg/dt" assumption breaks down, inside actively switching regions, not simply near a threshold voltage. The companion slow-term read, $C_{\text{apparent}} = dI/dt|_k - A_{\text{apparent}}\,dV/dt|_k$, is recoverable from the same pair wherever dg/dt is genuinely non-negligible (median 12% relative error there), but is a meaningless near-zero-denominator artifact in the roughly 75% of kinks that sit in an already-frozen-g region.

4.5 Curvature-based threshold estimators

Unlike the three relations above, this one is motivated analytically but not exact, and we are explicit about that distinction. Near a SET transition, I depends on g only through the affine blends $K_a(g), K_i(g)$, so by the chain rule d^2I/dt^2 picks up a term proportional to d^2g/dt^2 on top of the purely fast (dV_A/dt-driven) contribution already covered by Eq. (10). Since $dg/dt = (1-g)S(V_A)$ with $S(V_A) = 2e^{-\eta_s v_s}\sinh(\eta_s V_A)$ growing rapidly once V_A exceeds roughly v_s, d^2g/dt^2 itself peaks near where the SET rate's own growth is fastest – which is correlated with, but not identical to, the crossing $V_A = v_s$, because the $(1-g)$ prefactor and the fast-term contribution both shift the location of the maximum away from v_s by an amount that depends on the whole free-parameter side, not on v_s alone. This is why the resulting detectors below are usable *indicators* but not substitutes for v_s, v_r themselves. Two such detectors locate an *apparent* v_s or v_r directly from the curve's own shape: the first positive peak of d^2I/dt^2 outside a margin around $V = 0$ (excluding a real but unrelated artifact from the instrument's own acquisition-rate schedule), and the peak of d^3I/dt^3 at or before the peak of d^2I/dt^2, which for v_r specifically nearly doubles the correlation with the true model value over the plain second-derivative detector (0.247 to 0.513 in a synthetic check). Both derivatives are smoothed on a uniform *time* grid, not a fixed sample count – real acquisition dt varies several-fold within a single window, and smoothing by sample count badly distorts a second or third time derivative under that kind of jitter. We stress what these detectors are *not*: they correlate only weakly to moderately with the true model v_s, v_r (0.42–0.45 at best, uncalibrated), because the apparent threshold they locate depends on the whole SET or RESET nonlinearity, not on v_s or v_r in isolation. An earlier version of this project used them as the *reported* parameter value and as a hard constraint fixed inside the joint optimizer; both uses failed outright (the fixed-threshold variant produced unphysical fits in essentially every window, since the internal voltage V_A often cannot reach the apparent threshold at all given realistic fast-side parameters). They are correct and useful only as an input *feature* alongside the raw curve, which is how they are used in the CNN+lsq method of Section 3.

4.6 A non-gradient alternative: the rule-based parameter extractor

Every method in Section 3 is, underneath, a gradient descent or a population search against a single joint cost function. The tooling released alongside this paper (Appendix A) also includes a sixth, categorically different procedure: a rule-based parameter extractor, labeled the *fuzzy extractor* in the viewer's own tab, the name it inherited from the version it evolved from. That earlier version genuinely worked by ±1-vote fuzzy-logic rules, one per parameter, combined and thresholded each round; it proved hard to calibrate and audit, and the mechanism described below is its successor, keeping the one-rule-per-parameter spirit but replacing the voting with the direct, per-parameter correction detailed next.

The current procedure never computes a joint Jacobian, a population, or even a single joint cost value: `rmse` is simulated and recorded purely as a reported diagnostic, by explicit design, and never decides whether a step is accepted or when the procedure stops. Instead, each of the ten free parameters (γ held fixed, g_0 pinned by continuity, as throughout this paper) has its own extraction procedure, mapping a (t, V, I) curve directly to a value in that parameter's own physical units. Several of these are exactly the relations of Section 4 applied literally, rather than used only as a GA lock or a CNN input feature: v_s, v_r from the curvature detectors of Section 4's last subsection, r_{on}, r_{off} from the near-V=0 relation of Eq. (12). One relation appears here for the first time: η_s, η_r are read from the full width at half maximum, in time, of smoothed $|dI/dt|$'s own peak at the transition – a narrower peak means faster switching, i.e. a larger η_s or η_r, a direct geometric reading rather than a fit.

At each visit, the parameter's own procedure is applied once to the real data and once to the model's own current simulated curve over the same window; the correction is driven by the difference between the two – additive, directly in volts, for v_s, v_r; multiplicative, in log space, for the rest – a genuine Newton-style correction in the parameter's own physical units, not a step along a joint gradient and not a vote. Parameters are visited one at a time, in a fixed order $(v_r, v_s, i_{on}, r_{on}, r_{off}, a_{on}, a_{off}, i_{off}, \eta_s, \eta_r)$, one correction per visit, for repeated full passes (up to 40) until a pass moves nothing. A real divergence found during development motivated capping each individual correction (to ±2%): an undamped multiplicative step could swing i_{off} by a factor of 50 in one update, which then corrupted r_{off}'s own near-V=0 extraction in turn (Eq. (12) reads r_{off} off a measurement that itself depends on i_{off}) – a small, self-contained instance of exactly the kind of parameter entanglement Section 6 documents at the whole-chain scale.

We report this method's mechanism and design rationale in detail because of its interpretability – every update is a one-line, physically readable statement (*"the data's transition is narrower than the model's own, so raise η_s"*), auditable without any optimization-theory background, which none of the five methods in Section 3 offer. We also ran it through the same 500-half-cycle Monte Carlo ground-truth benchmark as those five (Section 5); the result is not a quantitative win – it is, if anything, the single clearest illustration in this paper of why the problem is hard, discussed alongside the other five methods' own results there rather than separately here.

5 Monte Carlo ground-truth comparison

Comparing extracted parameters against each other, or against previous runs of the same method, cannot distinguish "converged to the correct answer" from "converged to the same wrong answer as last time." We instead built a test with a known, exact ground truth: 500 half-cycles (250 full cycles) of parameters, taken directly from a real chained GA fit, were used to forward-simulate noise-free synthetic current over each half-cycle's own real (t, V) excitation. Every extraction strategy was then re-run on this synthetic data, blind to the generating parameters, and the recovered values

Table 2: Correlation between extracted and true parameter values, 500-half-cycle noise-free Monte Carlo test. Bold marks the best method per row.

parameter	lsq	GA	directed GA	GA+lsq	CNN+lsq	rule-based
v_s	**0.31**	0.18	0.14	0.16	0.21	−0.02
η_s	0.65	0.60	0.72	**0.89**	0.04	−0.08
r_{on}	0.64	**0.79**	0.72	0.78	0.48	−0.11
a_{on}	**0.45**	0.21	0.30	0.27	0.35	−0.12
i_{on}	0.24	0.06	0.00	0.04	**0.78**	−0.06
v_r	0.54	0.45	0.39	0.45	**0.70**	−0.07
η_r	**0.63**	0.13	0.18	0.07	0.18	0.09
r_{off}	0.27	0.28	**0.47**	0.41	0.10	−0.02
a_{off}	**0.79**	0.57	0.56	0.54	0.33	0.05
i_{off}	0.44	0.57	0.44	**0.64**	−0.07	−0.06
average	**0.50**	0.38	0.39	0.42	0.31	−0.04

compared against the known truth.

Table 2 and Figure 5 give the full result; Figure 6 shows the underlying scatter for lsq on four representative parameters. Five findings stand out.

First, plain lsq has the best *average* correlation of the six methods (0.50), despite being by far the cheapest. Population-based and hybrid methods do not buy accuracy here; GA+lsq, at roughly 17 times lsq's cost, trails it by 0.08 on average. We take this as evidence that the difficulty is not local-optimum trapping, which population methods are designed to escape, but something that survives a global search as well.

Second, no method reaches even moderate correlation on i_{on} except CNN+lsq (0.78 against a next-best of 0.24), and no method does well on v_s (0.14–0.31 across the board). These two are the project's most persistently non-identifiable parameters, and we return to why in Section 6.

Third, the CNN surrogate's error profile is qualitatively different from every optimizer-based method, not merely worse or better: it is the best method on i_{on} and v_r by a wide margin, and the worst on η_s, a_{off}, i_{off}, dragging its average below every other method. A network trained on realistic parameter *combinations*, rather than searching the current window in isolation, appears to encode a learned prior over where real devices actually sit along the (r_{on}, a_{on}, i_{on}) ridge described in Section 6 – information no local or population search run on a single curve has access to, but exactly the kind of information a single, fixed excitation waveform cannot by itself supply (Section 6). This was a single run of a flat, fully-connected network with one random seed and no architecture search; whether a properly convolutional, larger, ensembled version closes the gap on the remaining parameters without losing the i_{on}/v_r gain is, at the time of writing, still an open, partially negative experiment (one round of scaling up moved v_r's correlation to roughly 0.52 under teacher-forcing, but did not establish a clean win end to end).

Fourth, comparing this table against the same methods' agreement with each other on *real* (noisy) data is itself informative: lsq and GA agree with each other on real data at correlations of only 0.01–0.26 across the twelve parameters, far below what Table 2 shows on noise-free synthetic data (0.06–0.79). Most of that gap is explained by real measurement noise destabilizing an already only weakly constrained fit, not by a disagreement between the methods' underlying cost surfaces – but the gap does not close to zero even with noise removed, which is the residual, genuine non-identifiability this paper is about.

Fifth, the rule-based extractor of Section 4.6 is the clearest illustration of this paper's whole

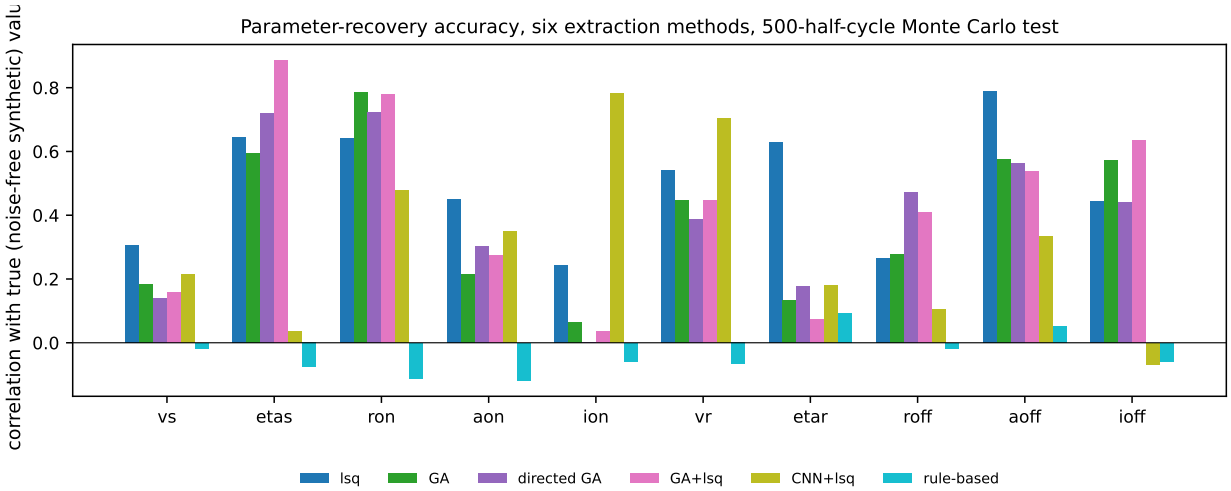


Figure 5: Correlation with the true, noise-free synthetic parameter value, all ten free parameters, six extraction methods. No method is uniformly best; i_{on} and v_s are weakly recovered by every method except, for i_{on}, the CNN surrogate. The rule-based extractor correlates weakly or negatively with truth on every parameter.

argument, not an outlier to it. We chain it cycle to cycle – bootstrapping the first cycle and warm-starting every subsequent one from the previous cycle's own output, the same continuity argument Section 2 makes for the other five methods, with a retry from a fresh bootstrap whenever a cycle's own warm start fails to switch – the same safeguard Section 3 already uses for the other five. Under that protocol, every one of its ten free parameters varies freely across the 250-cycle run, yet every one correlates weakly or negatively with truth (-0.12 to 0.09), giving the worst average of all six methods (-0.04). We read this as the same kind of evidence the bound-pinning pattern in Section 6 already provides for lsq/GA's a_{on} (Figure 7): a method built entirely differently from the other five – no gradient, no population, no learned prior, every update a one-line physical rule – still fails to recover the same parameters the other five struggle with, which is independent evidence that the difficulty documented throughout this paper is a property of the data, not of any one optimization strategy.

6 Why fitting this model is hard

The Monte Carlo test isolates the problem from measurement noise; it does not yet explain it. We checked four candidate explanations directly, rather than assume any of them.

A single, repeated excitation waveform. Across the entire usable dataset (cycles 2 through 16603), $V_{max} = 1.100\,\mathrm{V}$, $V_{min} = -1.400\,\mathrm{V}$, and the sample count per cycle are exactly identical, with zero variance, cycle after cycle – confirmed directly, not assumed. There is no hidden amplitude sweep or rate variation anywhere in the dataset to exploit. By the identifiability arguments of Chis and Balsa-Canto [8] and White et al. [9], a single fixed excitation, repeated, cannot in general carry enough independent information to uniquely constrain every parameter of a ten-to-twelve-parameter nonlinear ODE, regardless of how the inverse problem is solved. We consider this the deepest root cause of the pattern in Table 2: the specific parameters that go unrecovered are consistent across all five methods, which is what this explanation predicts and an optimizer-quality

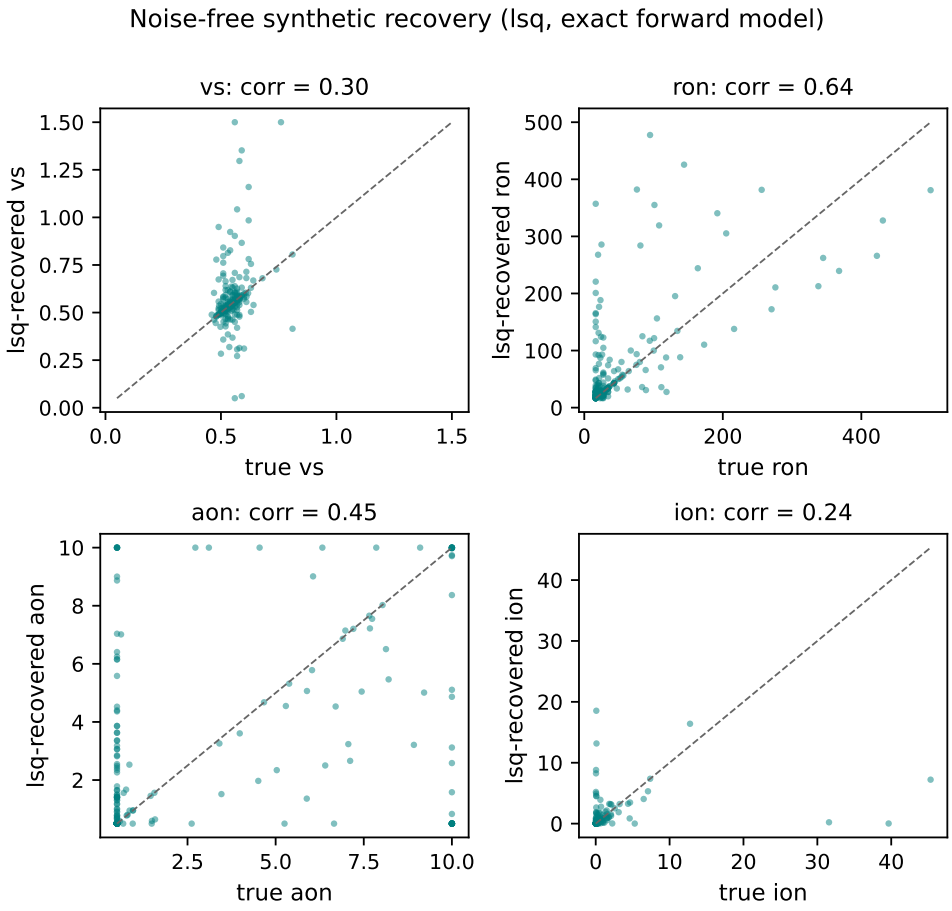


Figure 6: True vs. lsq-recovered value, noise-free synthetic data, four representative SET-side parameters. r_{on} clusters reasonably around the identity line; a_{on} and i_{on} do not.

explanation does not.

The γ/η_r ridge. Already described in Section 3 as a fitting-procedure fix: RESET's rate depends on $\eta_r\, g^{\gamma}$, and within a single half-cycle g does not range far enough for γ and η_r to be told apart. This is a genuine algebraic near-degeneracy, confirmed by scanning γ with η_r re-optimized at each point and finding the data matched to within a few percent across roughly a threefold range of γ. It is specific to the model's own g^{γ} term, and it is fully addressed by holding γ fixed; it is not the explanation for the remaining parameters.

Bound pinning on a_{on}. Figure 7 shows the distribution of a_{on} across the Monte Carlo ground-truth set (itself sourced from real GA-chain fits): 64% of values sit at the lower box bound, 14% at the upper one. An optimizer pinned against a box constraint, rather than converging to an interior optimum, is close to the textbook signature of a flat direction in the cost function – the search rides it to the edge of the box and stops there for lack of anything to distinguish points along it. A direct check of why: within real windows, the product $K_a V_A$ – the argument of the model's sinh nonlinearity – sits at a median of 0.3–0.8 and occasionally exceeds 2, which is too large for the small-signal linearization $\sinh(x) \approx x$ to hold cleanly across the whole window, yet the data still

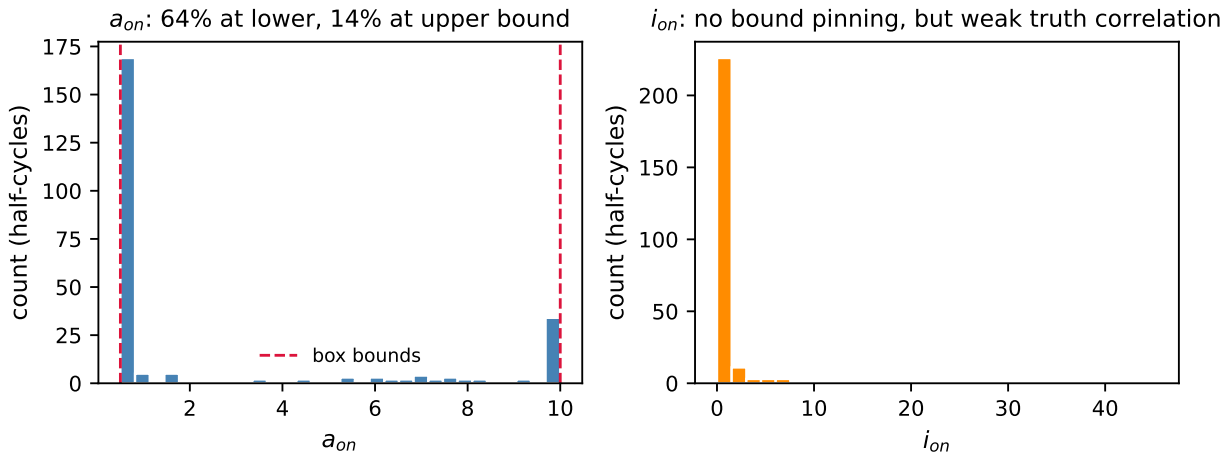


Figure 7: Distribution of a_{on} and i_{on} across the Monte Carlo ground-truth set (itself sourced from real fits). a_{on} shows the classic bimodal pile-up at its own box bounds that a flat-direction optimizer produces; i_{on} shows no such pile-up, yet is still poorly recovered (Table 2), which is why we tested the pooled hypothesis below rather than stopping at this figure.

does not pin a_{on} down on its own. We read this as consistent with, not contradicting, the waveform-diversity explanation above: the window explores enough of the nonlinearity to be sensitive to *some* combination of a_{on} and i_{on}, but not enough, under one fixed excitation, to separate the two.

A pooled-fitting test, reported here for the first time. The most natural fix to try, given the bound-pinning result, is to ask whether a_{on} and i_{on} are in fact device-level material constants that should not vary from cycle to cycle at all, with the apparent per-window scatter in Table 2 and Figure 7 being an artifact of fitting them independently under a weak per-window constraint. We tested this directly on twelve real SET half-cycle windows, spread across cycles 3000–8500 of the actual measured data (not synthetic), comparing two fits: each window's own fully free 11-parameter fit, against one joint least-squares problem over all twelve windows simultaneously, with a_{on} and i_{on} forced to a single shared value and every other parameter still free per window. If the hypothesis were correct, the pooled fit should cost little.

It does not. The pooled fit is worse than the free fit by a factor of 14.6 on average, and – this is the part that rules the hypothesis out rather than merely weakening it – *uniformly* worse across all twelve windows (Figure 8), not concentrated in a few outliers. The per-window free-fit values of a_{on} and i_{on} span a wide range (0.50–8.7 and 0.098–15.4 respectively) that a shared constant cannot absorb by adjusting the other nine free parameters per window; no other parameter can compensate for a wrong nonlinearity shape. a_{on} and i_{on} are therefore not simply masked constants: whatever each window's best-fit value actually reflects, it moves window to window in a way a single global value cannot capture, even though, per Section 4 and the single-window flat-direction evidence above, it is also only weakly determined *within* any one window. The most consistent reading of both results together is that this is a genuinely sloppy model in Chis and Balsa-Canto's sense: a real, nearly flat direction exists in each window's own cost surface, but its location shifts with the other, genuinely cycle-varying parameters (plausibly real device drift in r_{on}, v_s, or the switching rate), so no fixed point along it is right for every window.

Put together, these four checks point to the same conclusion from different directions: the DMM's own equations are not over-parameterized in a way that a cleverer reparameterization

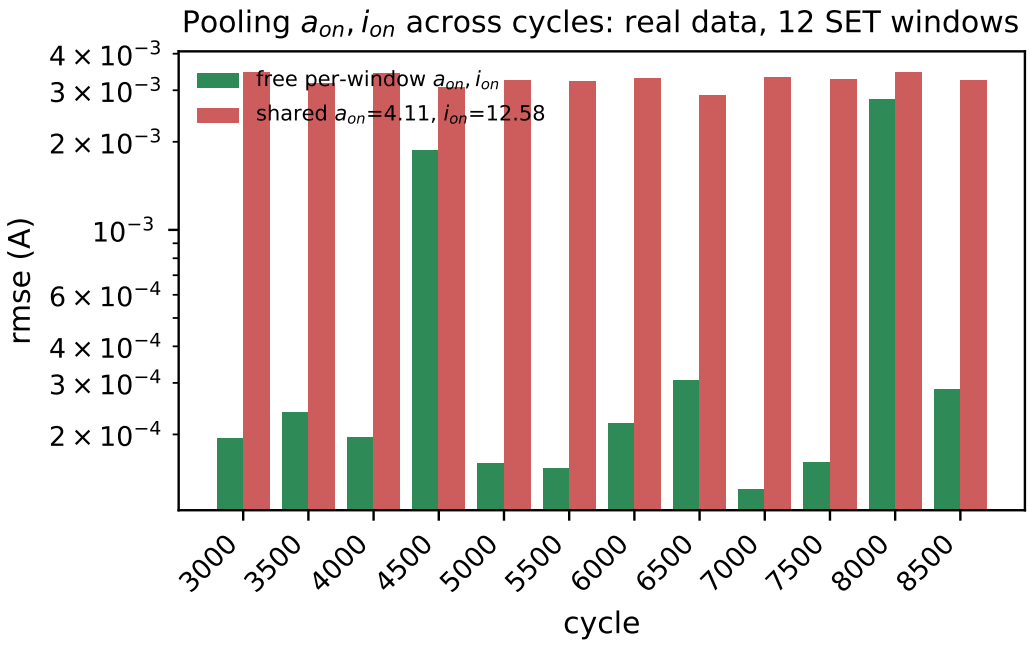


Figure 8: Fit rmse, real data, twelve SET half-cycle windows (cycles 3000–8500): each window fit fully independently, against one joint fit sharing a single a_{on}, i_{on} pair across all twelve windows. The pooled fit is worse in every single window, not just on average, ruling out "shared device constant hidden by noise" as an explanation.

could fix for free (the one genuine algebraic redundancy we found, γ against η_r, is already accounted for), and the remaining weak identifiability of v_s, a_{on}, i_{on}, η_r is a property of this specific, fixed-waveform dataset interacting with a twelve-parameter nonlinear ODE, not a property any of the five optimization strategies we tried could have fixed by searching harder or more cleverly.

7 Conclusions

The DMM reproduces this dataset's measured switching current accurately: a median peak-current error of 0.7% on the SET side and 1.7% on the RESET side, across a 14 604-cycle production chain fit, with no catastrophic derailment once the γ/η_r ridge and the window-reachability failure mode of Section 3 are addressed. Accurate current reproduction and accurate parameter recovery are not the same claim, however, and our central result is that they diverge substantially for a specific subset of this model's twelve parameters – most persistently v_s, a_{on}, i_{on}, and η_r – across every one of five genuinely different extraction strategies, spanning gradient descent, population search, a hybrid of the two, a method built explicitly around the local Jacobian's principal directions, and a learned surrogate trained on realistic synthetic data.

We do not think this reflects badly on the DMM specifically. The same pattern – parameters described in the literature as "fitting parameters without direct physical meaning," or reported saturating at search-space bounds during a recent independent CNN-based extraction effort on a different model entirely [5] – recurs across essentially every ReRAM compact model whose extraction procedure is documented in enough detail to check. What this dataset adds, beyond the general argument, is a direct, falsifiable demonstration that the obvious fix – treat the weakly-identified

parameters as shared constants across cycles – does not work here, tested rather than assumed.

Two practical consequences follow. First, any downstream use of this project's per-cycle parameter trajectories – aging or degradation analysis, for instance, which depends on treating cycle-to-cycle parameter *drift* as a physical signal – should not be taken at face value for v_s, a_{on}, i_{on}, or η_r without independent corroboration; the trajectory shape for these may reflect which point along a flat direction the optimizer happened to land on, not a real physical change. r_{on}, a_{off}, and (for GA-family methods) η_s are recovered considerably better and are safer to trust for this purpose. Second, since the root cause we identify is a property of the experiment, not the optimizer, the only extraction-side remedy left is an experimental one: varying the sweep amplitude or rate across a subset of cycles, specifically to give the fitting procedure information this dataset's single repeated waveform structurally cannot supply. We have not attempted this, since it requires new measurements rather than new analysis, but it follows directly from the argument above and we would recommend it as the next step for anyone extending this work.

A The rolling-parameter analyzer

Every per-cycle figure in this paper (Figures 3, 7, and the trajectory view used while diagnosing the pin-escape and rmse-spike failure modes of Section 3) was produced with the aid of a tool we release alongside this paper, [12]:

https://github.com/rpicos-uib/rolling-parameter-analyzer

It is a Tkinter desktop viewer over a chained-fit CSV (one row per half-cycle, the twelve DMM parameters plus fit `rmse`), built incrementally over the course of this project specifically because aggregate statistics repeatedly hid real problems that only a direct, per-cycle visual check caught – the chain-damping, curve-inspector, and window-reachability issues of Section 3 were all found this way, not from a summary number. It reads its input CSV on demand, so it doubles as a live monitor for a still-running extraction job, and provides six views:

1. **Evolution vs. cycle** – any subset of the twelve parameters, plus `rmse`, each independently rescaled to $[-1, 1]$ so parameters on very different scales (e.g. $i_{off} \sim 10^{-4}$ against $r_{on} \sim 50$) can be read off one shared axis against cycle number. A cycle slider opens a popup overlaying the real measured curve against the model's own forward simulation, for the selected cycle and its neighbors, each with its own fitted parameters.

2. **Parameter vs. parameter** – a scatter of any two parameters, same cycle or lagged by one, with a regression line and its R^2.

3. **Distribution** – a histogram of one parameter (or `rmse`) with any combination of Gaussian, Lognormal, Gamma, Exponential, Logistic, Skew-normal, or Weibull fits overlaid, plain or as a k-component mixture.

4. **Parameter table** – a live, scrollable table of all twelve parameters against cycles, centered on a chosen cycle.

5. **Curve inspector** – actual-vs-model current and I-V for a chosen cycle. This view required a specific fix during this project: a chain-fitted cycle has two different parameter sets in play, one before and one after its own RESET-side update, and a single merged simulation call across the whole cycle silently applies the wrong, later set to the earlier portion of the curve

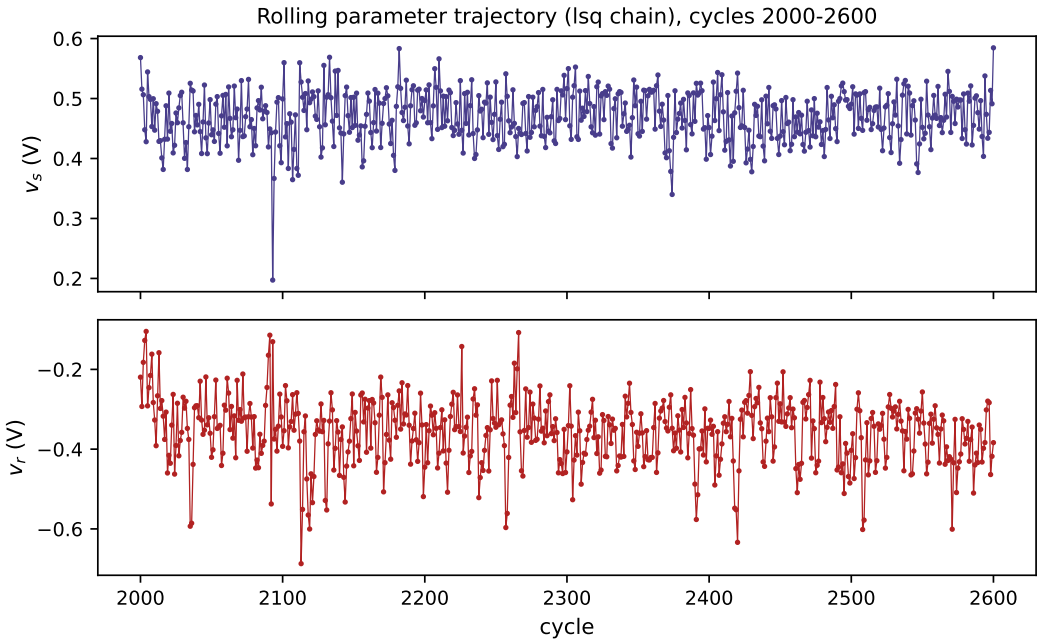


Figure 9: The kind of trajectory the "Evolution vs. cycle" view is built to show: v_s and v_r as extracted by the production lsq chain, cycles 2000–2600. Consistent with the identifiability findings of Section 6, this trajectory should be read as a noisy, only weakly constrained signal rather than a precise physical record of cycle-to-cycle drift.

– confirmed directly as the cause of a 20–30% spurious reconstruction error that had nothing to do with the underlying extraction, which was correct throughout.

6. **Fuzzy extractor** – runs the project's interpretable, per-parameter iterative extraction procedure (Section 4.6) live on one chosen cycle, showing each individual parameter update (which parameter, its own data-vs-model error, old and new value) rather than only a converged endpoint, as an alternative, steppable lens alongside the joint least-squares fit.

Beyond the viewer itself, the repository also includes the fitting CLI behind Section 3 (`fit_chain_cli.py`, exposing four of the paper's five extraction methods – lsq, GA, directed GA, and GA+lsq – through one documented `--help`) and the Monte Carlo ground-truth generator and comparison script behind Section 5 (`mc_param_recovery.py`, `compare_vs_truth.py`), all under the MIT license. A small synthetic demo dataset is committed alongside them so every command the repository's own README documents – generate, fit, validate, inspect – runs end to end on a fresh clone, with no dependency on the real measurement data [11], which is not included. The paper's fifth method (the CNN surrogate) was a single exploratory pass, as Sections 3 and 5 already note, and its research-stage code is likewise not part of this release.

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